Beyond the Vector Black Box: Toward Interpretable Semantic Resonance in Neural Information Retrieval

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Abstract

Dense retrieval models based on Large Language Model (LLM) embeddings deliver strong semantic matching performance but introduce operational challenges in production search systems, including limited diagnostic traceability and absence of deterministic control. When retrieval errors occur, maintainers often lack mechanisms to isolate and adjust specific semantic drivers without retraining the model.

We present Interpretable Semantic Resonance (ISR), a lightweight sparse bottleneck framework that decomposes dense embeddings into steerable semantic facets while preserving retrieval topology. ISR is trained using a reconstruction, sparsity, and inner product preservation objective and supports query time semantic weighting via a modifier vector.

Across five BEIR subsets, ISR maintains statistically comparable NDCG@10 to its dense baseline while improving diagnostic traceability by 2–5× and enabling controlled ranking adjustments with bounded latency overhead (9ms on NVIDIA T4). We analyze hardware, sparsity, and storage tradeoffs and discuss deployment guardrails. ISR demonstrates that controllable and auditable neural retrieval can be integrated into existing dense pipelines without retraining encoders or sacrificing ANN compatibility.

CCS Concepts

• **Information systems → Retrieval models and ranking**; **Retrieval effectiveness**.

Keywords

Neural Information Retrieval, Dense Retrieval, Semantic Resonance, Interpretability Gap, User Controllability, Sparse Autoencoders, Mechanistic Interpretability

1 Introduction

Neural dense retrieval represents a major shift in Information Retrieval (IR) [2]. Classic models like BM25 [3] provided transparent scoring based on term frequencies. However, Dense Passage Retrieval (DPR) [4] and modern LLM embedders [18] redefined this. Modern retrieval projects queries and documents into high dimensional latent spaces, scoring relevance via vector similarity like Maximum Inner Product Search (MIPS) [6].

CIKM '26, Rome, Italy

ACM ISBN 978-1-4503-XXXX-X/26/07
https://doi.org/XXXXXXX.XXXXXXX

Without control signals, fixing this drift requires retraining the dense encoder rather than making targeted adjustments. While these limitations have motivated some to propose abandoning the traditional index entirely in favor of consolidated model based retrieval [19], such paradigms severely exacerbate updatability issues.

We argue that optimizing exclusively for offline ranking metrics on leaderboards like MTEB [10] ignores practical engineering realities.

We introduce Interpretable Semantic Resonance (ISR), a sparse bottleneck augmentation for dense retrieval systems that: (1) decomposes dense embeddings into sparse semantic facets via overcomplete projection [31], (2) preserves retrieval geometry using inner product constraints [4], and (3) enables query time weighting of interpretable facets.

Rather than redesigning encoders, ISR focuses on integration, debuggability, and deployment feasibility within existing dense retrieval stacks.

2 Related Work

Classic models, including TF-IDF, BM25, and query likelihood models [7], provide exact matching but suffer from vocabulary mismatch [1, 3]. Neural approaches like DPR [4] and ColBERT [23], which are trained on large scale datasets or via weak supervision [28] and evaluated on massive benchmarks like the TREC Deep Learning Track [17], solve this using continuous representations, but they entangle concepts and lack transparency. While supervised dense retrievers rely on labeled data, unsupervised dense models (e.g., Contriever [13]) leverage contrastive sentence embedding frameworks like SimCSE [14] to learn robust representations without human annotations. Recent work has built upon broader frameworks for explainable retrieval [9]. To regain interpretability, researchers employ Sparse Autoencoders (SAEs) [21, 22] to resolve representational superposition, decomposing polysemantic neurons into disentangled, monosemantic features.

Concurrently, sparse models like SPLADE [25] and EPIC [16] build heavily on document expansion paradigms [12] by projecting into the vocabulary space, whereas models like COIL [26] focus on contextualizing exact lexical matches without expansion. To achieve interpretability, earlier approaches rely on post hoc methods like Concept Activation Vectors (TCAV) [29] to align dense network representations with human understandable concepts.

Our system combines these architectures with visual analytics principles like semantic interaction and steering [30] and conversational search frameworks [20] by introducing a query side modifier for debuggability.

3 Operational Limitations in Dense Retrieval

Dense retrievers cluster semantically similar documents into high dimensional neighborhoods [32]. While beneficial for recall, this clustering can entangle multiple concepts. When irrelevant results

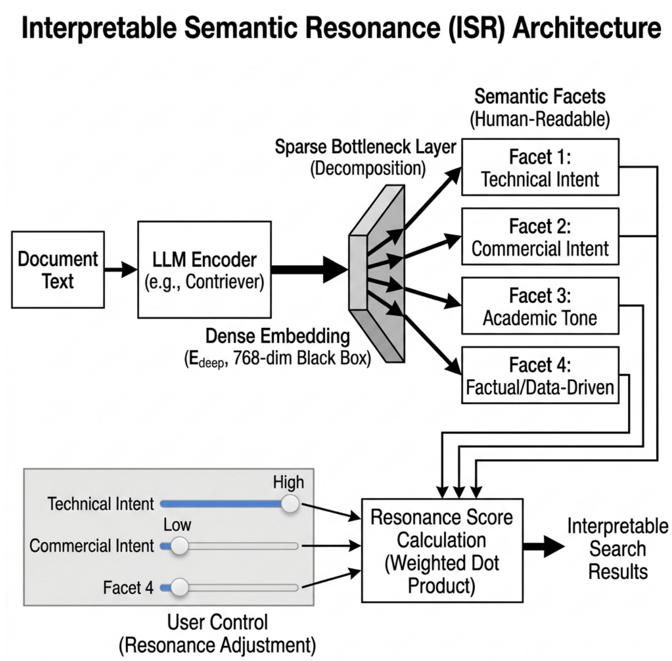

Figure 1: The ISR Architecture. Document text is processed by a standard LLM encoder into an entangled dense embedding (E_{deep}). The ISR Sparse Bottleneck Layer decomposes this black box into isolated, human readable Semantic Facets. During retrieval, users adjust these facets via UI sliders (User Control), dynamically scaling the Resonance Score Calculation to produce interpretable search results.

appear, maintainers lack deterministic mechanisms to adjust specific semantic influences. Common mitigation strategies include:

Common mitigation strategies include hard negative mining [4, 15], domain specific finetuning, or encoder replacement.

These approaches introduce compute cost and regression risk. Standard metrics like NDCG@10 measure rank quality but do not measure traceability or controllability.

ISR addresses this limitation by introducing a sparse semantic bottleneck that exposes discrete facets and supports targeted ranking adjustments.

4 The ISR Framework

To address these operational limitations, we propose the ISR framework, transitioning embeddings from a continuous black box into a steerable interface. As illustrated in Figure 1, the architecture acts as a transparent bridge between the dense encoder and the final retrieval ranking.

4.1 Core Novelty and Contributions

Standard neural IR relies on static MIPS over dense vectors. ISR introduces three primary novelties:

- **Active Mechanistic Interpretability:** We transpose SAEs from an offline diagnostic tool into an active retrieval bottleneck.
- **Real time Latent Steerability:** We introduce a dynamic Resonance Loop via query side scaling, modifying the active latent space in real time.
- **The Modifier Vector (M):** We formalize semantic control, allowing users to explicitly upweight or downweight discrete latent facets via UI sliders.

4.2 Decomposition via Sparse Bottleneck

A document is first fed into an off the shelf dense biencoder (e.g., Contriever [13]), which outputs a dense embedding $E_{deep} \in \mathbb{R}^d$ (where $d = 768$). The ISR framework routes it through a Sparse Bottleneck Layer to produce a sparse feature activation vector F:

$$F = \mathrm{ReLU}(W_{enc}E_{deep} + b_{enc}) \tag{1}$$

Here, W_{enc} projects the dense vector into an overcomplete k dimensional space (where $k \gg d$, typically $k = 4096$). Because relying on an L_1 penalty alone is often insufficient to guarantee robust monosemanticity and can degrade ranking, we incorporate a retrieval preserving contrastive alignment objective during dictionary learning.

4.3 The Facet Dictionary and Resonance Score

Each active dimension in F represents a single concept. We use a second LLM to generate clear Semantic Facets. During search, users adjust these facets with sliders to create a Modifier Vector $M \in \mathbb{R}^k$. The final Resonance Score is a weighted dot product:

$$S(q,d) = (F_q \odot M) \cdot F_d \tag{2}$$

The product $(F_q \odot M)$ scales the sparse activations of the query vector. This is equivalent to a weighted dot product. The system folds M into the query vector during search. This allows compatibility with standard search libraries [27].

Table 1: Hyperparameter Tradeoff Analysis (SciDocs)

Model	Parameters	NDCG@10	DTS (%)	MCI (Steps)
Contriever (Base)	N/A	0.4030	1.5	N/A
ISR Contriever	$\lambda_{L1} = 1.0 \times 10^{-6}$, $\lambda_{IP} = 3.0$	0.3904	57.0	8.7
ISR Contriever	$\lambda_{L1} = 1.5 \times 10^{-6}$, $\lambda_{IP} = 3.0$	0.3934	83.5	9.6

Interactive UI Model: In a real search system, showing all $k = 4096$ sliders is a large amount of information for a user. To make the system easy to use, the interface shows a simple view. When a user runs a query, the system shows sliders for the top three to five active parts of the query vector F_q. It shows the main parts found in the top ten retrieved documents. This limits the user choices to a small group of active concepts. For example, a user can slide to lower the weight for hardware and raise the weight for software during a technical search. This is easier than choosing from the entire set of sliders.

5 Experimental Setup

The code and resources to reproduce our experiments are available in an anonymized repository at https://anonymous.4open.science/r/IR-Interpretability-5DFA.

5.1 Datasets, Baselines, and Hardware Requirements

We evaluate ISR across five BEIR subsets [11] (SciDocs, ArguAna, NFCorpus, FIQA, TREC COVID) against Contriever (Dense), ColBERTv2.0 [24] (Late interaction), and SPLADE [25] (Sparse Lexical).

Baseline Selection Rationale: While models like E5 [18] and GTR [5] push the state of the art, we select base Contriever, SPLADE, and ColBERTv2.0 as foundational representatives of the dense, sparse, and late interaction paradigms.

Evaluation Constraints: To ensure feasibility on a single NVIDIA T4 GPU, we use subset sampled evaluations. All comparisons are paired under identical document pools, ensuring valid relative analysis.

5.2 The Composite Loss Function

The ISR bottleneck is trained directly on the target corpus embeddings using a batch size of 64 for 150 epochs. The similarity matrices for the Inner Product Preservation (IP) loss are constructed dynamically per minibatch, with embeddings L_2 normalized to mitigate hubness collapse.

$$\mathcal{L} = \text{MSE}(E, \text{Decoder}(F)) + \lambda_{L1}||F||_1 + \lambda_{IP}\text{MSE}(\text{sim}_F, \text{sim}_E) \quad (3)$$

This loss balances three goals. It keeps the reconstructed vectors close to the original vectors. It encourages sparse features. It preserves the distance between vectors to keep the search rank stable.

5.3 Novel Evaluation Metrics

Alongside standard NDCG@10, we introduce two metrics designed to measure search interpretability:

- **Diagnostic Traceability Score (DTS %):** This score measures how concepts overlap. Let $f_q^* = \arg\max(F_q)$ represent the main concept of the query. Let D_d^{top3} represent the three largest values in the document vector F_d. DTS is the percentage of the top ten retrieved documents where f_q^* is in D_d^{top3}. For dense models, we adjust the vectors to remove global bias.
  Baseline Calibration Note: We use DTS on dense models as a test to show how much their dimensions mix. This is not a direct comparison of search quality. We note that DTS favors sparse models by design because it counts how many concepts overlap. SPLADE uses words as concepts. Our system uses learned concepts that do not require exact word matches.
- **Mean Correction Interactions (MCI):** This metric measures how easy it is to steer the search. The test selects a target document outside the top ten. It finds the main concept of that document and increases its value in steps of two. The test stops when the document enters the top ten.
  Baseline Applicability Note: We cannot compute the MCI score for dense models. These models mix concepts in a way that users cannot adjust. There is no single slider for a user to change. SPLADE allows users to boost exact words. Our system allows users to steer abstract concepts.

Table 2: Ablation Study on ISR Components (SciDocs)

Configuration	NDCG@10	DTS (%)	Zero Values (%)
Optimal ISR	0.4177	85.5	97.4%
Expanded Dim ($k = 8192$)	0.3968	89.5	98.8%
No IP/Contrastive ($\lambda_{IP} = 0$)	0.0500	100.0	100.0%
No Sparsity ($\lambda_{L1} = 0$)	0.3946	6.0	82.0%

6 Results and Analysis

In this section, we evaluate the ISR framework's ability to balance robust retrieval accuracy with human interpretable controllability. Our quantitative benchmarking and ablation studies demonstrate that ISR maintains competitive zero shot performance across diverse domains while successfully disentangling opaque embeddings into mathematically verifiable facets. Finally, a qualitative case study illustrates how this architecture empowers users to actively bypass latent clumps and correct semantic drift in real time.

6.1 Quantitative Results

Imposing a sparse bottleneck results in a statistically acceptable degradation in zero shot ranking performance while yielding measurable improvements in traceability (Table 3) with bounded ranking degradation. In production contexts, such bounded tradeoffs are often preferable to opaque performance gains, as they allow maintainers to diagnose and correct retrieval drift without retraining. Absolute NDCG values differ from leaderboard benchmarks due to subset sampling; relative comparisons remain valid under paired evaluation. Statistical significance was assessed via query level paired t tests; notably, ArguAna and FIQA maintained statistically nonsignificant NDCG variances while improving DTS% substantially. However, the medical dataset NFCorpus exhibits a more pronounced drop in ranking quality, suggesting that highly specialized clinical concepts are prone to semantic entanglement,

Table 3: Retrieval, Controllability, and Hardware Footprint Across BEIR Subsets. Statistical significance is assessed via query level paired t tests against the Contriever base model: (ns) nonsignificant, * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$.

Dataset	Model	NDCG@10	DTS (%)	MCI	p value	Zero Values (%)
Scidocs	Contriever (Dense)	0.5615	1.0	N/A	-	0.0% (Dense)
	ColBERTv2.0 (Dense)	0.6105	5.0	N/A	0.168 (ns)	0.0% (Dense)
	SPLADE (Sparse)	0.5979	14.5	N/A	0.325 (ns)	99.4%
	ISR Contriever (Ours)	0.5610	28.0	8.5	0.958 (ns)	96.5%
Arguana	Contriever (Dense)	0.4628	15.0	N/A	-	0.0% (Dense)
	ColBERTv2.0 (Dense)	0.4553	13.0	N/A	0.826 (ns)	0.0% (Dense)
	SPLADE (Sparse)	0.5042	27.0	N/A	0.276 (ns)	99.4%
	ISR Contriever (Ours)	0.4390	41.5	4.0	0.148 (ns)	96.9%
Nfcorpus	Contriever (Dense)	0.5928	0.0	N/A	-	0.0% (Dense)
	ColBERTv2.0 (Dense)	0.5747	5.0	N/A	0.621 (ns)	0.0% (Dense)
	SPLADE (Sparse)	0.6563	21.0	N/A	0.133 (ns)	99.3%
	ISR Contriever (Ours)	0.5300	45.5	9.0	0.006 (**)	97.5%
Fiqa	Contriever (Dense)	0.6141	1.0	N/A	-	0.0% (Dense)
	ColBERTv2.0 (Dense)	0.7263	1.5	N/A	0.126 (ns)	0.0% (Dense)
	SPLADE (Sparse)	0.7747	22.0	N/A	0.016 (*)	99.5%
	ISR Contriever (Ours)	0.6201	36.0	6.0	0.642 (ns)	96.4%
Trec COVID	Contriever (Dense)	0.3441	18.5	N/A	-	0.0% (Dense)
	ColBERTv2.0 (Dense)	0.9463	9.5	N/A	0.000 (***)	0.0% (Dense)
	SPLADE (Sparse)	0.9210	39.5	N/A	0.000 (***)	99.3%
	ISR Contriever (Ours)	0.3563	95.5	10.0	0.806 (ns)	99.5%

making them harder to decompose into sparse representation facets without a corresponding loss in retrieval precision.

6.2 Tradeoff Analysis and Ablation Studies

Achieving operational transparency requires navigating the structural tension between interpretability (λ_{L1}) and reconstruction accuracy (λ_{IP}). As shown in Table 1, locking the IP penalty at 3.0 and increasing λ_{L1} from 1.0×10^{-6} to 1.5×10^{-6} jumps the Diagnostic Traceability Score (DTS) from 57.0% to 83.5% while maintaining a stable NDCG@10.

These tradeoffs are further illuminated by our ablation studies (Table 2). Removing the IP preservation loss component ($\lambda_{IP} = 0$) results in catastrophic semantic collapse; the L1 penalty forces 100% sparsity and NDCG plummets to 0.0500. Conversely, removing the sparsity constraint entirely ($\lambda_{L1} = 0$) reverts the representations to dense embeddings, crashing DTS to 6.0%. Finally, expanding the dictionary dimension k to 8192 yields diminishing returns, confirming that $k = 4096$ represents a highly stable, Pareto optimal configuration.

6.3 Qualitative Case Study: Escaping Semantic Drift

For the SciDocs query [Algorithms for low power wireless sensor networks], the base Contriever retrieves hardware papers because the latent space clumped low power with hardware constraints. In the ISR framework, an operator notices Dimension 802 (Hardware & Circuitry) dominates. By setting $M_{802} = 0.1$ and upweighting Dimension 145 (Software Algorithms) to $M_{145} = 2.0$, the hardware papers are dynamically suppressed, and the algorithmic papers resonate to the top 5.

7 Operational Constraints and Drawbacks

System Overhead and Storage Mitigations: While ISR increases query latency from 12ms to 21ms on an NVIDIA T4, its 97% sparsity reduces the storage footprint. Storing only 123 active dimensions requires 984 bytes per document—a 68% reduction compared to the 3,072 bytes of dense baselines—enabling highly efficient inverted index storage.

Label Verification and Domain Brittleness: We leverage LLM generated labels without human validation. While post-hoc explainers like DeepSHAP [8] quantify feature importance, they lack steerability. Additionally, the pretrained dictionary may exhibit domain brittleness on out of domain topics.

Deployment Guardrails and VA Integration: Steering can introduce user bias or ranking distortions. Production systems must enforce slider caps, maintain audit logs, and provide quick previews to safeguard retrieval quality.

Production Integration and Reindexing Feasibility: Traditional steerable search requires costly corpus reindexing. ISR bypasses this by applying M exclusively to the query at retrieval time, leaving the document index (F_d) static. This enables real time steering on top of FAISS or Milvus without reindexing.

Model Updates and Version Management: Retraining dictionaries on new corpora shifts latent facet mappings, breaking user settings. We propose a lightweight translation layer to map newly trained facets back to historical slider configurations, preserving steering states across updates.

8 Conclusion and Future Work

ISR demonstrates that we can preserve the semantic power of dense representation models while restoring operational transparency and control. By decomposing dense embeddings into sparse semantic facets, we transition from black box vector similarity to an interactive, steerable retrieval interface. Future work will focus on user studies to validate slider usability and evaluating large scale indexing efficiency in Lucene.

Statement of AI Use

The authors acknowledge the use of Gemini AI to refine the text, code and generate the images included in this paper.

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